

PROJECT 2 — REVENUE LEAKAGE ANALYSIS

```
/* =====  
SECTION 1 — DATA QUALITY & VALIDATION  
===== */
```

-- 1.1 Total record count

```
SELECT COUNT(*) AS totalrecords  
FROM ecommerce;
```

	123 totalrecords
1	1,200

-- 1.2 Check for duplicate order IDs

```
SELECT orderid,  
       COUNT(*) AS duplicatecount  
FROM ecommerce  
GROUP BY orderid  
HAVING COUNT(*) > 1;
```

A-Z orderid	123 duplicatecount

-- 1.3 Check for NULL values in critical columns

```
SELECT  
    SUM(CASE WHEN orderid IS NULL THEN 1 ELSE 0 END) AS nullorderid,  
    SUM(CASE WHEN date IS NULL THEN 1 ELSE 0 END) AS nulldate,
```

```

SUM(CASE WHEN customerid IS NULL THEN 1 ELSE 0 END) AS nullcustomer,
SUM(CASE WHEN product IS NULL THEN 1 ELSE 0 END) AS nullproduct,
SUM(CASE WHEN orderstatus IS NULL THEN 1 ELSE 0 END) AS nullstatus,
SUM(CASE WHEN totalprice IS NULL THEN 1 ELSE 0 END) AS nullprice,
SUM(CASE WHEN couponcode IS NULL THEN 1 ELSE 0 END) AS nullcoupon
FROM ecommerce;

```

--

	123 nullorderid	123 nulldate	123 nullcustomer	123 nullproduct	123 nullstatus	123 nullprice	123 nullcoupon
1	0	0	0	0	0	0	0

-- 1.4 Verify all order statuses are expected values

```

SELECT orderstatus,
       COUNT(*) AS recordcount
FROM ecommerce
GROUP BY orderstatus
ORDER BY recordcount DESC;

```

	AZ orderstatus	123 recordcount
1	Cancelled	250
2	Returned	247
3	Pending	237
4	Shipped	235
5	Delivered	231

-- 1.5 Date range validation

```

SELECT
  MIN(date) AS earliestorder,
  MAX(date) AS latestorder,
  COUNT(DISTINCT date) AS uniquedates

```

FROM ecommerce;

		earliestorder		latestorder	123	uniquedates
--	1	2023-01-01	2025-06-30	671		

/* =====

SECTION 2 — REVENUE HEALTH OVERVIEW

Business Question: What is the total scale of the revenue leakage problem?

===== */

-- 2.1 Revenue breakdown by order status

SELECT

orderstatus,

COUNT(*) AS totalorders,

ROUND(SUM(totalprice), 2) AS totalrevenue,

ROUND(AVG(totalprice), 2) AS avgordervalue,

ROUND(

COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2

) AS pctoforders,

ROUND(

SUM(totalprice) * 100.0 / SUM(SUM(totalprice)) OVER (), 2

) AS pctofrevenue

FROM ecommerce

GROUP BY orderstatus

ORDER BY totalrevenue DESC;

	AZ orderstatus	123 totalorders	123 totalrevenue	123 avgordervalue	123 pctoforders	123 pctofrevenue
1	Cancelled	250	276,396.21	1,105.58	20.83	21.85
2	Pending	237	256,328.15	1,081.55	19.75	20.27
3	Shipped	235	246,159.58	1,047.49	19.58	19.46
4	Returned	247	243,277.7	984.93	20.58	19.24
5	Delivered	231	242,600.32	1,050.22	19.25	19.18

-- 2.2 Lost revenue summary (Cancelled + Returned)

```
SELECT
COUNT(*)                                AS lostorders,
ROUND(SUM(totalprice), 2)                 AS totallostrevenue,
ROUND(AVG(totalprice), 2)                 AS avglostperorder,
ROUND(
SUM(totalprice) * 100.0 /
(SELECT SUM(totalprice) FROM ecommerce), 2
)                                           AS pctoftotalrevenue
FROM ecommerce
WHERE orderstatus IN ('Cancelled', 'Returned');
```

--

	123 lostorders	123 totallostrevenue	123 avglostperorder	123 pctoftotalrevenue
1	497	519,673.91	1,045.62	41.09

-- 2.3 Leakage vs healthy revenue comparison

```
SELECT
CASE
WHEN orderstatus IN ('Cancelled', 'Returned') THEN 'Lost Revenue'
WHEN orderstatus = 'Delivered' THEN 'Confirmed Revenue'
ELSE 'In-Progress Revenue'
END                                           AS revenuecategory,
COUNT(*)                                   AS orders,
ROUND(SUM(totalprice), 2)                   AS revenue
```

FROM ecommerce

GROUP BY revenuecategory

ORDER BY revenue **DESC**;

	A-Z revenuecategory ▼	123 orders ▼	123 revenue ▼
1	Lost Revenue	497	519,673.91
2	In-Progress Revenue	472	502,487.73
3	Confirmed Revenue	231	242,600.32

```
/* =====  
SECTION 3 — LOST REVENUE BY PRODUCT  
===== */
```

-- 3.1 Full product-level loss analysis

SELECT

product,

COUNT(*) **AS** totalorders,

SUM(**CASE WHEN** orderstatus = 'Cancelled' **THEN** 1 **ELSE** 0 **END**) **AS** cancelledorders,

SUM(**CASE WHEN** orderstatus = 'Returned' **THEN** 1 **ELSE** 0 **END**) **AS** returnedorders,

SUM(**CASE WHEN** orderstatus **IN** ('Cancelled','Returned') **THEN** 1 **ELSE** 0 **END**)

AS totallostorders,

ROUND(

SUM(**CASE WHEN** orderstatus **IN** ('Cancelled','Returned')

THEN totalprice **ELSE** 0 **END**), 2

) **AS** lostrevenue,

ROUND(

SUM(**CASE WHEN** orderstatus **IN** ('Cancelled','Returned') **THEN** 1 **ELSE** 0 **END**)

* 100.0 / **COUNT**(*), 1

```

)
AS lossratepct,
ROUND(AVG(
CASE WHEN orderstatus IN ('Cancelled','Returned')
THEN totalprice END), 2
)
AS avglostordervalue
FROM ecommerce
GROUP BY product
ORDER BY lossratepct DESC;

```

--

	A2 product	I23 totalorders	I23 cancelledorders	I23 returnedorders	I23 totallostorders	I23 lostrevenue	I23 lossratepct	I23 avglostordervalue
1	Monitor	163	35	36	71	77,582.04	43.6	1,092.7
2	Tablet	179	34	43	77	83,155.82	43	1,079.95
3	Laptop	173	35	39	74	83,416.02	42.8	1,127.24
4	Chair	178	45	28	73	75,299.14	41	1,031.5
5	Printer	181	35	38	73	74,614.25	40.3	1,022.11
6	Phone	156	31	31	62	57,187.46	39.7	922.38
7	Desk	170	35	32	67	68,419.18	39.4	1,021.18

-- 3.2 Product cancellation vs return breakdown

```

SELECT
product,
ROUND(
SUM(CASE WHEN orderstatus = 'Cancelled' THEN 1 ELSE 0 END)
* 100.0 / COUNT(*), 1
) AS cancellationratepct,
ROUND(
SUM(CASE WHEN orderstatus = 'Returned' THEN 1 ELSE 0 END)
* 100.0 / COUNT(*), 1
) AS returnratepct
FROM ecommerce
GROUP BY product
ORDER BY cancellationratepct DESC;

```

	AZ product ▼	123 cancellationratepct ▼	123 returnratepct ▼
1	Chair	25.3	15.7
2	Monitor	21.5	22.1
3	Desk	20.6	18.8
4	Laptop	20.2	22.5
5	Phone	19.9	19.9
6	Printer	19.3	21
7	Tablet	19	24

/* =====

SECTION 4 — LOST REVENUE BY PAYMENT METHOD

===== */

-- 4.1 Payment method risk analysis

SELECT

paymentmethod,

COUNT(*) AS totalorders,

SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')

THEN 1 ELSE 0 END) AS lostorders,

ROUND(

SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')

THEN totalprice ELSE 0 END), 2

) AS lostrevenue,

ROUND(

SUM(CASE WHEN orderstatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END)

* 100.0 / COUNT(*), 1

) AS lossratepct,

ROUND(

SUM(CASE WHEN orderstatus = 'Delivered' THEN 1 ELSE 0 END)

```

        * 100.0 / COUNT(*), 1
    ) AS deliverysuccesspct

FROM ecommerce

GROUP BY paymentmethod

ORDER BY lossratepct DESC;

```

--

	A2 paymentmethod	123 totalorders	123 lostorders	123 lostrevenue	123 lossratepct	123 deliverysuccesspct
1	Gift Card	230	102	110,084.75	44.3	16.1
2	Credit Card	234	103	117,733.25	44	18.4
3	Cash	246	106	112,052.85	43.1	18.3
4	Debit Card	232	95	87,633.56	40.9	18.5
5	Online	258	91	92,169.5	35.3	24.4

-- 4.2 Payment method — revenue successfully delivered

```

SELECT

    paymentmethod,

    ROUND(SUM(CASE WHEN orderstatus = 'Delivered'

        THEN totalprice ELSE 0 END), 2) AS confirmedrevenue,

    ROUND(SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')

        THEN totalprice ELSE 0 END), 2) AS lostrevenue,

    ROUND(

        SUM(CASE WHEN orderstatus = 'Delivered' THEN totalprice ELSE 0 END)

        * 100.0 / SUM(totalprice), 1

    ) AS revenuerecoverypct

FROM ecommerce

GROUP BY paymentmethod

ORDER BY revenuerecoverypct DESC;

```


--

	AZ paymentmethod	123 confirmedrevenue	123 lostrevenue	123 revenuerecoverypct
1	Online	67,119.65	92,169.5	25.6
2	Debit Card	45,885.14	87,633.56	19.7
3	Credit Card	46,054.96	117,733.25	17.5
4	Gift Card	40,682.88	110,084.75	16.5
5	Cash	42,857.69	112,052.85	16.5

/* =====

SECTION 5 — MARKETING CHANNEL QUALITY ANALYSIS

===== */

-- 5.1 Channel-level loss and order quality

SELECT

referralsource,

COUNT(*) AS totalorders,

SUM(CASE WHEN orderstatus = 'Delivered' THEN 1 ELSE 0 END) AS delivered,

SUM(CASE WHEN orderstatus = 'Cancelled' THEN 1 ELSE 0 END) AS cancelled,

SUM(CASE WHEN orderstatus = 'Returned' THEN 1 ELSE 0 END) AS returned,

ROUND(

SUM(CASE WHEN orderstatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END)

* 100.0 / COUNT(*), 1

) AS lossratepct,

ROUND(

SUM(CASE WHEN orderstatus = 'Delivered' THEN 1 ELSE 0 END)

* 100.0 / COUNT(*), 1

) AS qualityscorepct,

```

ROUND(
    SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')
        THEN totalprice ELSE 0 END), 2
    ) AS lostrevenue
FROM ecommerce
GROUP BY referralsource
ORDER BY lossratepct DESC;

```

--

	A-Z referralsource	123 totalorders	123 delivered	123 cancelled	123 returned	123 lossratepct	123 qualityscorepct	123 lostrevenue
1	Email	250	60	59	49	43.2	24	114,021.54
2	Facebook	228	42	42	56	43	18.4	115,294.12
3	Referral	222	33	50	44	42.3	14.9	95,222.24
4	Google	241	44	58	43	41.9	18.3	95,356.06
5	Instagram	259	52	41	55	37.1	20.1	99,779.95

-- 5.2 Revenue value per channel

```

SELECT
    referralsource,
    ROUND(AVG(totalprice), 2) AS avgordervalue,
    ROUND(SUM(totalprice), 2) AS totalgmV,
    ROUND(SUM(CASE WHEN orderstatus = 'Delivered'
        THEN totalprice ELSE 0 END), 2) AS netdeliveredrevenue,
    ROUND(
        SUM(CASE WHEN orderstatus = 'Delivered' THEN totalprice ELSE 0 END)
        * 100.0 / SUM(totalprice), 1
    ) AS revenueefficiencypct
FROM ecommerce
GROUP BY referralsource
ORDER BY revenueefficiencypct DESC;

```

--

	A-Z referralsource	123 avgordervalue	123 totalgmV	123 netdeliveredrevenue	123 revenueefficiencypct
1	Email	1,047.23	261,808.55	64,553.52	24.7
2	Facebook	1,098.29	250,410.9	52,149.38	20.8
3	Instagram	1,062.88	275,285.45	51,299.56	18.6
4	Google	1,039.18	250,441.48	41,406.25	16.5
5	Referral	1,021.69	226,815.58	33,191.61	14.6

/* =====

SECTION 6 — COUPON CODE ABUSE DETECTION

===== */

-- 6.1 Coupon code performance and abuse rate

SELECT

COALESCE(couponcode, 'NO COUPON') **AS** couponcode,

COUNT(*) **AS** totalorders,

SUM(**CASE WHEN** orderstatus **IN** ('Cancelled','Returned')

THEN 1 **ELSE** 0 **END**) **AS** lostorders,

ROUND(

SUM(**CASE WHEN** orderstatus **IN** ('Cancelled','Returned') **THEN** 1 **ELSE** 0 **END**)

* 100.0 / **COUNT**(*), 1

) **AS** lossratepct,

ROUND(

SUM(**CASE WHEN** orderstatus = 'Returned' **THEN** 1 **ELSE** 0 **END**)

* 100.0 / **COUNT**(*), 1

) **AS** returnratepct,

ROUND(

SUM(**CASE WHEN** orderstatus = 'Cancelled' **THEN** 1 **ELSE** 0 **END**)

* 100.0 / **COUNT**(*), 1

) **AS** cancelratepct,

ROUND(

SUM(**CASE WHEN** orderstatus **IN** ('Cancelled','Returned')

THEN totalprice **ELSE** 0 **END**), 2

) **AS** discountedlostrevenue

FROM ecommerce

GROUP BY couponcode

ORDER BY lossratepct **DESC**;

--

	AZ couponcode	123 totalorders	123 lostorders	123 lossratepct	123 returnratepct	123 cancelratepct	123 discountedlostrevenue
1	WINTER15	292	130	44.5	21.6	22.9	141,855.23
2		309	134	43.4	24.6	18.8	137,057.12
3	FREESHIP	313	128	40.9	19.5	21.4	138,929.1
4	SAVE10	286	105	36.7	16.4	20.3	101,832.46

-- 6.2 Coupon vs no-coupon order completion quality

SELECT

CASE WHEN couponcode **IS NOT NULL THEN** 'Coupon Used' **ELSE** 'No Coupon' **END**

AS couponusage,

COUNT(*) **AS** totalorders,

ROUND(**AVG**(totalprice), 2) **AS** avgordervalue,

ROUND(
SUM(**CASE WHEN** orderstatus **IN** ('Cancelled','Returned') **THEN** 1 **ELSE** 0 **END**)
* 100.0 / **COUNT**(*), 1

) **AS** lossratepct,

ROUND(
SUM(**CASE WHEN** orderstatus = 'Delivered' **THEN** 1 **ELSE** 0 **END**)
* 100.0 / **COUNT**(*), 1

) **AS** deliveryratepct

FROM ecommerce

GROUP BY couponusage

ORDER BY lossratepct **DESC**;

--

	AZ couponusage	123 totalorders	123 avgordervalue	123 lossratepct	123 deliveryratepct
1	Coupon Used	1,200	1,053.97	41.4	19.3

```

/* =====
SECTION 7 — MONTHLY CANCELLATION TREND (2023–2025)
===== */

```

-- 7.1 Monthly revenue and loss trend

```

SELECT
    DATE_TRUNC('month', date)::DATE AS ordermonth,
    COUNT(*) AS totalorders,
    ROUND(SUM(totalprice), 2) AS totalrevenue,
    SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')
        THEN 1 ELSE 0 END) AS lostorders,
    ROUND(
        SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')
            THEN totalprice ELSE 0 END), 2
    ) AS lostrevenue,
    ROUND(
        SUM(CASE WHEN orderstatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*), 1
    ) AS lossratepct
FROM ecommerce
GROUP BY ordermonth
ORDER BY ordermonth;
--

```

	ordermonth	123 totalorders	123 totalrevenue	123 lostorders	123 lostrevenue	123 lossratepct
1	2023-01-01	47	56,685.75	18	22,698.34	38.3
2	2023-02-01	37	40,117.66	20	22,656.99	54.1
3	2023-03-01	43	48,609.37	23	23,950.6	53.5
4	2023-04-01	31	27,751.71	11	8,905.23	35.5
5	2023-05-01	49	63,836.84	20	23,489.14	40.8
6	2023-06-01	45	49,500.19	21	24,871.1	46.7
7	2023-07-01	44	42,820.66	12	8,578.75	27.3
8	2023-08-01	51	54,352.14	24	24,799.68	47.1
9	2023-09-01	29	29,526.67	14	15,586.82	48.3
10	2023-10-01	47	52,607.85	14	12,284.1	29.8
11	2023-11-01	44	42,820.66	12	8,578.75	27.3

-- 7.2 Year-over-year leakage comparison

```
SELECT
    EXTRACT(YEAR FROM date) AS orderyear,
    COUNT(*) AS totalorders,
    SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')
        THEN 1 ELSE 0 END) AS lostorders,
    ROUND(
        SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')
            THEN totalprice ELSE 0 END), 2
    ) AS lostrevenue,
    ROUND(
        SUM(CASE WHEN orderstatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*), 1
    ) AS lossratepct
FROM ecommerce
GROUP BY orderyear
ORDER BY orderyear;
```

--

	123 orderyear	123 totalorders	123 lostorders	123 lostrevenue	123 lossratepct
1	2,023	510	209	222,095.51	41
2	2,024	459	180	179,895.94	39.2
3	2,025	231	108	117,682.46	46.8

-- 7.3 Quarterly performance overview

```
SELECT
    EXTRACT(YEAR FROM date) AS year,
    EXTRACT(QUARTER FROM date) AS quarter,
    CONCAT('Q', EXTRACT(QUARTER FROM date)::INT,
```

```

        '-', EXTRACT(YEAR FROM date)::INT) AS period,
COUNT(*) AS totalorders,
ROUND(SUM(totalprice), 2) AS totalrevenue,
ROUND(
    SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')
        THEN totalprice ELSE 0 END), 2
) AS lostrevenue,
ROUND(
    SUM(CASE WHEN orderstatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 1
) AS lossratepct
FROM ecommerce
GROUP BY year, quarter, period
ORDER BY year, quarter;
--

```

	123 year	123 quarter	AZ period	123 totalorders	123 totalrevenue	123 lostrevenue	123 lossratepct
1	2,023	1	Q1-2023	127	145,412.78	69,305.93	48
2	2,023	2	Q2-2023	125	141,088.74	57,265.47	41.6
3	2,023	3	Q3-2023	124	126,699.47	48,965.25	40.3
4	2,023	4	Q4-2023	134	139,442.25	46,558.86	34.3
5	2,024	1	Q1-2024	100	111,468.55	38,496.67	35
6	2,024	2	Q2-2024	137	145,590.79	61,100.93	43.8
7	2,024	3	Q3-2024	115	114,750.03	49,042.03	43.5
8	2,024	4	Q4-2024	107	108,426.5	31,256.31	32.7
9	2,025	1	Q1-2025	113	103,617.61	51,523.06	42.5
10	2,025	2	Q2-2025	118	128,265.24	66,159.4	50.8

```

/* =====
SECTION 7.4 — COHORT ANALYSIS: CHANNEL RETENTION & CANCELLATION BEHAVIOR
===== */

```

-- 7.4a Cohort analysis: Customer acquisition channel vs cancellation behavior

```

WITH customerfirstsource AS (
    SELECT

```

```

customerid,
referralsource,
MIN(date) AS firstorderdate,
MAX(date) AS lastorderdate,
COUNT(*) AS totalorders
FROM ecommerce
GROUP BY customerid, referralsource
)
SELECT
cfs.referralsource,
COUNT(DISTINCT cfs.customerid) AS uniquecustomers,
SUM(cfs.totalorders) AS totalordersfromcohort,
ROUND(AVG(cfs.totalorders), 2) AS avgorderspercustomer,
ROUND(
COUNT(DISTINCT CASE WHEN o.orderstatus IN ('Cancelled','Returned')
THEN o.orderid END)
* 100.0 / COUNT(DISTINCT o.orderid), 1
) AS cohortlossratepct,
ROUND(
SUM(CASE WHEN o.orderstatus IN ('Cancelled','Returned')
THEN o.totalprice ELSE 0 END), 2
) AS cohortlostrevenue
FROM customerfirstsource cfs
LEFT JOIN ecommerce o ON cfs.customerid = o.customerid
GROUP BY cfs.referralsource
ORDER BY cohortlossratepct DESC;

```


--

	AZ.referralsource	l23.uniquecustomers	l23.totalordersfromcohort	l23.avgorderspercustomer	l23.cohortlossratepct	l23.cohortlostrevenue
1	Email	250	254	1	42.9	114,370.04
2	Referral	221	226	1.01	42.9	99,801.33
3	Facebook	228	234	1	42.7	118,184.9
4	Google	241	243	1	41.6	95,356.06
5	Instagram	257	265	1.02	37.2	101,285.41

-- 7.4b Deep dive: Instagram vs Google cohort comparison

WITH channelcohorts AS (

SELECT

CASE WHEN referralsource IN ('Instagram') THEN 'Instagram'

WHEN referralsource IN ('Google') THEN 'Google'

ELSE 'Other' END AS channelgroup,

orderid,

customerid,

orderstatus,

totalprice,

date

FROM ecommerce

WHERE referralsource IN ('Instagram', 'Google')

)

SELECT

channelgroup,

COUNT(DISTINCT customerid) AS uniquecustomers,

COUNT(orderid) AS totalorders,

ROUND(

COUNT(CASE WHEN orderstatus = 'Delivered' THEN 1 END)

* 100.0 / COUNT(orderid), 1

) AS deliveryratepct,

ROUND(

COUNT(CASE WHEN orderstatus = 'Cancelled' THEN 1 END)

* 100.0 / COUNT(orderid), 1

```

) AS cancellationratepct,

ROUND(

    COUNT(CASE WHEN orderstatus = 'Returned' THEN 1 END)

    * 100.0 / COUNT(orderid), 1

) AS returnratepct,

ROUND(

    SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')

        THEN totalprice ELSE 0 END), 2

) AS lostrevenue,

ROUND(AVG(totalprice), 2) AS avgordervalue

FROM channelcohorts

GROUP BY channelgroup

ORDER BY cancellationratepct DESC;

```

	AZ channelgroup	123 uniquecustomers	123 totalorders	123 deliveryratepct	123 cancellationratepct	123 returnratepct	123 lostrevenue	123 avgordervalue
1	Google	241	241	18.3	24.1	17.8	95,356.06	1,039.18
2	Instagram	257	259	20.1	15.8	21.2	99,779.95	1,062.88

-- 7.4c Repeat purchase behavior by acquisition channel

```

WITH firstorderchannel AS (

    SELECT

        customerid,

        referralsource,

        MIN(date) AS firstorderdate

    FROM ecommerce

    GROUP BY customerid, referralsource

)

SELECT

    foc.referralsource,

    COUNT(DISTINCT foc.customerid) AS customersacquired,

    COUNT(DISTINCT CASE WHEN o.date > foc.firstorderdate

```

```

        THEN o.customerid END) AS customersrepeatpurchased,
ROUND(
    COUNT(DISTINCT CASE WHEN o.date > foc.firstorderdate
        THEN o.customerid END)
    * 100.0 / COUNT(DISTINCT foc.customerid), 1
) AS repeatpurchaseratepct,
ROUND(
    AVG(CASE WHEN o.date > foc.firstorderdate
        AND o.orderstatus IN ('Cancelled','Returned')
        THEN 1 ELSE 0 END) * 100, 1
) AS repeatorderlossratepct
FROM firstorderchannel foc
LEFT JOIN ecommerce o ON foc.customerid = o.customerid
GROUP BY foc.referralsource
ORDER BY repeatpurchaseratepct DESC;

```

--

	A2 referralsource	123 customersacquired	123 customersrepeatpurchased	123 repeatpurchaseratepct	123 repeatorderlossratepct
1	Instagram	257	4	1.6	0.4
2	Facebook	228	3	1.3	0.4
3	Referral	221	2	0.9	0.4
4	Email	250	2	0.8	0
5	Google	241	0	0	0

/* =====

SECTION 8 — ORDER FUNNEL CONVERSION RATE

===== */

-- 8.1 Full funnel with conversion rates

```

WITH funnel AS (
    SELECT
        SUM(CASE WHEN orderstatus IN ('Pending','Shipped','Delivered','Cancelled','Returned') THEN 1
        ELSE 0 END) AS placed,

```

```

SUM(CASE WHEN orderstatus IN ('Shipped','Delivered') THEN 1 ELSE 0 END) AS shipped,
SUM(CASE WHEN orderstatus = 'Delivered' THEN 1 ELSE 0 END) AS delivered,
SUM(CASE WHEN orderstatus = 'Cancelled' THEN 1 ELSE 0 END) AS cancelled,
SUM(CASE WHEN orderstatus = 'Returned' THEN 1 ELSE 0 END) AS returned
FROM ecommerce
)
SELECT
placed AS ordersplaced,
shipped AS ordersshipped,
delivered AS ordersdelivered,
cancelled AS orderscancelled,
returned AS ordersreturned,
ROUND(shipped * 100.0 / placed, 1) AS shipmentratepct,
ROUND(delivered * 100.0 / placed, 1) AS deliverysuccesspct,
ROUND(cancelled * 100.0 / placed, 1) AS cancellationratepct,
ROUND(returned * 100.0 / placed, 1) AS returnratepct
FROM funnel;

```

--

	123 ordersplac	123 ordersshipped	123 ordersdelivered	123 orderscancelled	123 ordersreturned	123 shipmentratepct	123 deliverysucc	123 cancellationrate	123 returnratepct
1	1,200	466	231	250	247	38.8	19.3	20.8	20.6

-- 8.2 Funnel by product

```

SELECT
product,
COUNT(*) AS total,
ROUND(SUM(CASE WHEN orderstatus = 'Delivered' THEN 1 ELSE 0 END)
* 100.0 / COUNT(*), 1) AS deliverypct,
ROUND(SUM(CASE WHEN orderstatus = 'Cancelled' THEN 1 ELSE 0 END)
* 100.0 / COUNT(*), 1) AS cancelpct,
ROUND(SUM(CASE WHEN orderstatus = 'Returned' THEN 1 ELSE 0 END)

```

```

* 100.0 / COUNT(*), 1) AS returnpct,

ROUND(SUM(CASE WHEN orderstatus = 'Pending' THEN 1 ELSE 0 END)

* 100.0 / COUNT(*), 1) AS pendingpct

FROM ecommerce

GROUP BY product

ORDER BY deliverypct DESC;

```

--

	A-Z product	123 total	123 deliverypct	123 cancelpct	123 returnpct	123 pendingpct
1	Phone	156	24.4	19.9	19.9	17.9
2	Laptop	173	23.1	20.2	22.5	17.3
3	Monitor	163	19	21.5	22.1	17.8
4	Desk	170	18.8	20.6	18.8	22.4
5	Chair	178	18.5	25.3	15.7	23
6	Printer	181	16	19.3	21	20.4
7	Tablet	179	15.6	19	24	19

```

/* =====

SECTION 9 — HIGH-RISK ORDER PROFILING (MULTI-DIMENSIONAL)

===== */

```

-- 9.1 Cross-dimensional risk: Product × Payment Method

```

SELECT

product,

paymentmethod,

COUNT(*) AS totalorders,

SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')

THEN 1 ELSE 0 END) AS lostorders,

ROUND(

SUM(CASE WHEN orderstatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END)

* 100.0 / COUNT(*), 1

) AS lossratepct,

ROUND(

```

```

SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')
    THEN totalprice ELSE 0 END), 2
) AS lostrevenue
FROM ecommerce
GROUP BY product, paymentmethod
HAVING COUNT(*) >= 10
ORDER BY lossratepct DESC
LIMIT 15;
--

```

	AZ product	AZ paymentmethod	123 totalorders	123 lostorders	123 lossratepct	123 lostrevenue
1	Tablet	Cash	35	19	54.3	20,494.15
2	Printer	Gift Card	29	15	51.7	18,761.69
3	Laptop	Gift Card	33	17	51.5	22,333.7
4	Printer	Debit Card	33	17	51.5	12,710.88
5	Chair	Gift Card	36	18	50	20,451.99
6	Monitor	Credit Card	34	17	50	23,196.38
7	Laptop	Cash	45	22	48.9	25,884.46
8	Desk	Debit Card	34	16	47.1	13,307.26
9	Phone	Credit Card	35	16	45.7	11,127.11
10	Tablet	Credit Card	31	14	45.2	20,440.75

-- 9.2 Cross-dimensional risk: Channel × Product

```

SELECT
    referralsource,
    product,
    COUNT(*) AS totalorders,
    ROUND(
        SUM(CASE WHEN orderstatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*), 1
    ) AS lossratepct,
    ROUND(
        SUM(CASE WHEN orderstatus IN ('Cancelled','Returned')
            THEN totalprice ELSE 0 END), 2
    ) AS lostrevenue

```

FROM ecommerce

GROUP BY referralsource, product

HAVING COUNT(*) >= 8

ORDER BY lossratepct **DESC**

LIMIT 15;

--

	AZ referralsource	AZ product	123 totalorders	123 lossratepct	123 lostrevenue
1	Email	Laptop	33	54.5	24,607.16
2	Referral	Chair	31	51.6	17,335.49
3	Facebook	Monitor	37	51.4	25,595.73
4	Google	Tablet	35	51.4	18,768.16
5	Facebook	Printer	36	50	15,034.21
6	Referral	Tablet	32	50	14,824.18
7	Google	Phone	39	48.7	15,488.39
8	Google	Laptop	33	48.5	15,309.08
9	Email	Phone	34	47.1	16,513.8
10	Facebook	Chair	37	45.9	17,444.09

-- 9.3 High-risk order flag

WITH productavg **AS** (

SELECT

product,

ROUND(**AVG**(totalprice), 2) **AS** avgproductprice

FROM ecommerce

WHERE orderstatus **IN** ('Cancelled','Returned')

GROUP BY product

)

SELECT

o.orderid,

o.date,

o.customerid,

o.product,

o.paymentmethod,

o.referralsource,

o.couponcode,

o.orderstatus,

o.totalprice,

pa.avgproductprice,

CASE

WHEN o.totalprice > pa.avgproductprice * 1.5 **THEN** 'HIGH RISK — Above 150% avg'

WHEN o.totalprice > pa.avgproductprice **THEN** 'MEDIUM RISK — Above avg'

ELSE 'STANDARD RISK'

END AS riskflag

FROM ecommerce o

JOIN productavg pa **ON** o.product = pa.product

WHERE o.orderstatus **IN** ('Cancelled','Returned')

ORDER BY o.totalprice **DESC**

LIMIT 20;

--

	AZ_orderid	date	AZ_customerid	AZ_product	AZ_paymentmethod	AZ_referralsource	AZ_couponcode	AZ_orderstatus	123_totalprice	123_avgproductprice
1	ORD201122	2023-06-07	C38840	Monitor	Online	Facebook		Returned	3,390.95	
2	ORD200469	2023-11-26	C13877	Chair	Cash	Facebook		Cancelled	3,384.9	
3	ORD200328	2023-02-28	C18404	Tablet	Online	Google	SAVE10	Cancelled	3,370.2	1
4	ORD200326	2024-07-01	C65986	Laptop	Gift Card	Facebook	SAVE10	Returned	3,352.4	1
5	ORD200527	2024-08-19	C27202	Chair	Credit Card	Referral	SAVE10	Cancelled	3,267.35	
6	ORD200768	2023-09-04	C35987	Tablet	Cash	Email	WINTER15	Cancelled	3,267.3	1
7	ORD200889	2024-10-27	C36725	Monitor	Credit Card	Email	WINTER15	Cancelled	3,253.6	
8	ORD200802	2024-02-17	C84331	Chair	Gift Card	Email	FREESHIP	Cancelled	3,223.2	
9	ORD200957	2023-01-12	C73575	Monitor	Cash	Facebook	WINTER15	Returned	3,219.45	
10	ORD200986	2024-06-10	C82205	Printer	Online	Referral	FREESHIP	Cancelled	2,315.15	1